CSE4203 — Principles of Software Architecture

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Building Height Compliance Platform

A multi-tier system for advisory building-height compliance assessment in Egypt

Milestone 1

Proposal and Architectural Requirements

Team name	BHCP
Milestone	M1 — Proposal & Architectural Requirements
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Team Members

Name	Student ID	Primary role in M1
Loay Waleed	231004836	Repository setup; system overview; stakeholder analysis
Ali Elsayed	231004649	System context diagram; functional requirements
Mohamed Ahmed Tohamy	231015004	Quality attribute scenarios; trade-off analysis
Ziad Mohamed Hamam	231006572	Constraints; ASRs; references; final assembly

Diagram source files are held in `/diagrams` in the repository.

Contribution Table

Member	Contribution to Milestone 1	Sections
Loay Waleed	Initialised the repository and the document build. Defined the system scope and problem statement; identified the stakeholder set and their architectural concerns. Coordinated the hand-off sequence between members.	Cover, 1, 2
Ali Elsayed	Produced the C4 Level 1 context diagram and its legend in Ti*k*Z; documented the five external integrations and the deployment split. Elicited and wrote the 15 user stories, applied MoSCoW prioritisation and selected the Milestone 3 prototype subset.	3, 4
Mohamed Ahmed Tohamy	Authored the six quality attribute scenarios in six-part form and derived their measurable response measures. Analysed the four competing quality attribute pairs and the resolution adopted for each.	5
Ziad Mohamed Hamam	Collected the technical, business and organisational constraints. Built the utility tree, ranked the architecturally significant requirements and wrote their justification. Compiled the IEEE reference list and performed final document assembly and proof-reading.	6, 7, References

All four members reviewed the complete document before submission and agree the distribution above is accurate.

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1 System Overview

The problem

Egyptian building regulations tie the permitted height of a building to the width of the street it fronts, with additional limits set by zone and by the regulations in force when the building was licensed [6]. Verifying whether a given building respects those limits currently requires a site visit by a qualified inspector with surveying equipment. The Ministry of Housing, Utilities and Urban Communities (MHUC) cannot staff that process at the scale of a growing city, so non-compliant construction is typically discovered late, after it is expensive to remedy.

The **Building Height Compliance Platform (BHCP)** addresses this from two directions at once. Building owners can measure their own property from the street using an ordinary smartphone, and receive an advisory indication of whether it appears to comply before a problem escalates. MHUC officers can screen an entire district from satellite data and obtain a ranked list of buildings worth visiting, rather than inspecting at random.

Who uses it

Three distinct roles, with deliberately asymmetric capability. A **building owner** may assess only buildings they have verified ownership of; they measure using phone sensors and photographs, and they cannot see other people's assessments. An **MHUC officer** has access to the satellite screening tier and the review queue, and is the only role permitted to act on a flagged case. A **system administrator** authors and versions the regulatory rule sets and manages role assignments, but issues no verdicts.

Why it is architecturally interesting

Four properties make this more than a CRUD application.

The system's output is a claim about someone's property, so correctness is a safety property, not a feature. A false positive is an accusation against a lawful building. The architecture must therefore propagate measurement uncertainty end to end and refuse to issue a verdict when the confidence interval overlaps the legal limit. This pushes a numerical concern — calibrated error propagation — into the core domain model, where it constrains the API, the database schema and the UI.

Capability varies by platform and cannot be abstracted away. Augmented-reality ranging is available on Android via ARCore and on iOS via ARKit, but browsers expose no visual-inertial odometry at all. Rather than degrading to the lowest common denominator, the architecture defines explicit measurement tiers with different accuracy envelopes, and the uncertainty attached to a result records which tier produced it.

The rules change independently of the software. Ministerial decrees amend height bands on their own schedule, and a verdict must be evaluated against the rules in force on the building's *licence* date, not today's. Regulation is therefore modelled as versioned, effective-dated data rather than code, and historical assessments must remain reproducible.

Measurement happens where the network is not. Owners stand in streets with poor connectivity, so capture must work fully offline and reconcile later — which puts availability and consistency in direct tension with the integrity requirement above.

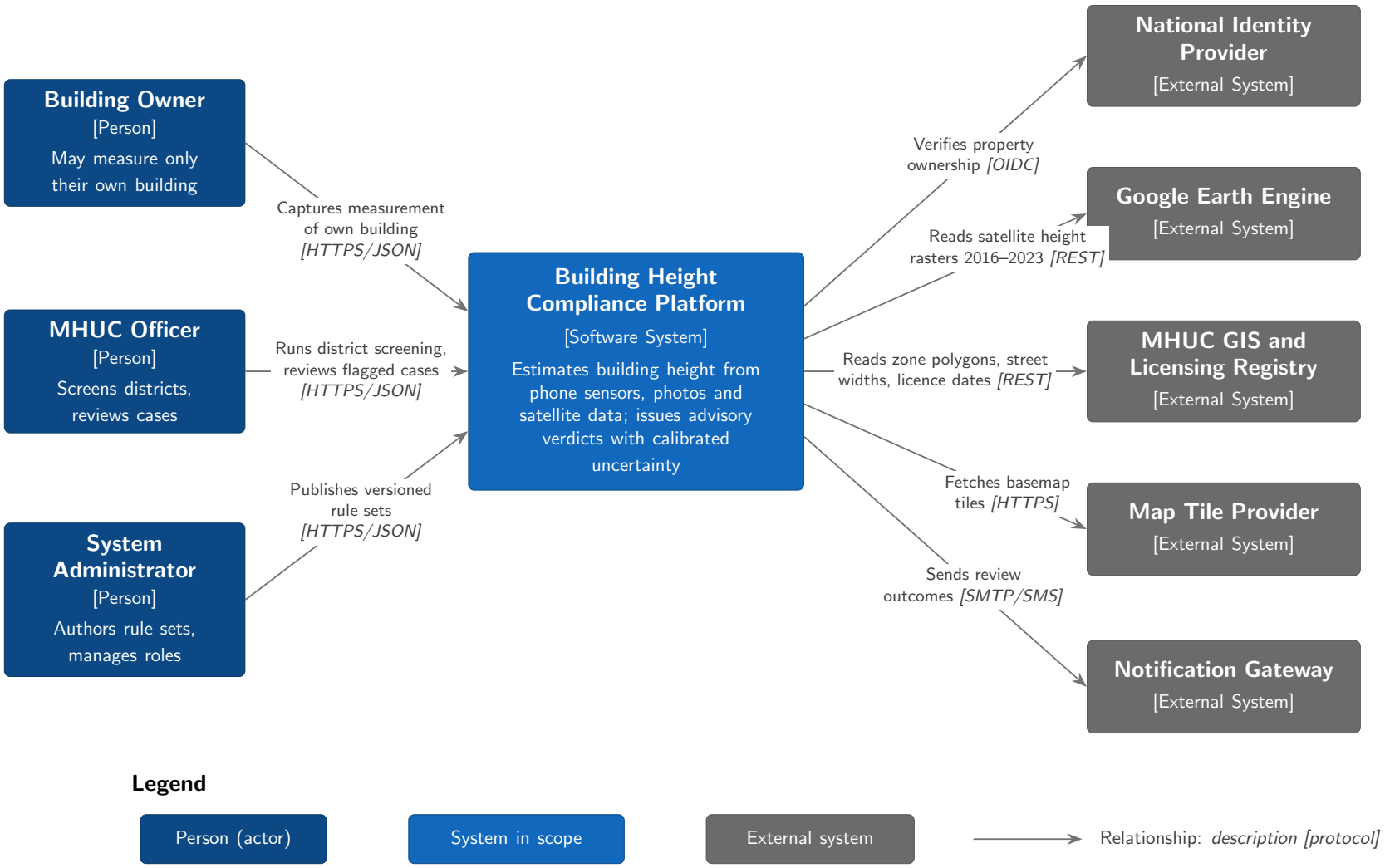
2 Stakeholders and Concerns

Table 2: Stakeholders and their principal architectural concerns

Stakeholder	Interest	Key concerns
Building owner	Primary end user	Wants a trustworthy answer without hiring a surveyor. Concerned that the result is accurate enough to rely on, that the measurement is quick and works without a signal, and that assessing their own property does not expose them to enforcement they did not invite.
MHUC officer	Regulatory user	Needs to prioritise a limited inspection budget. Concerned with screening throughput over a whole district, the false-positive rate of flagged candidates, and being able to see which decree and which street width a verdict rests on before acting.
System administrator	Operator of the rule base	Concerned that a new decree can be published quickly and safely, that effective dates are respected, and that changing a rule never silently rewrites verdicts already issued.
Development team	Builders	Concerned with modifiability and testability: shared measurement logic must not be reimplemented per platform, and the numerical core must be verifiable against ground-truth data independently of any UI.
Operations	Deployment and running	Concerned with running inside a free or low-cost cloud tier, bounded satellite API quota consumption, observability of the inference path, and safe schema evolution.
MHUC legal affairs	Regulator	Concerned that the system is clearly advisory and cannot be mistaken for a legal determination, that every verdict is auditable to a cited decree, and that assessments are retained with their original inputs.
Uninvolved residents	Affected third party	Not users, but affected. Concerned that their faces, vehicles and homes are not captured, stored or published, and that the platform cannot be turned into a tool for neighbour disputes.
Course supervisor	Project sponsor	Concerned with delivery inside the academic calendar, demonstrable architectural reasoning, and a Milestone 3 prototype that genuinely conforms to the documented design.

3 System Context

Figure 1 shows the C4 Level 1 system context [4]. The platform is drawn as a single box; everything outside it is integrated with rather than built by the team.



C4 Model Level 1 (System Context). Everything outside the blue box is beyond the team's control: it is integrated with, not built.

Figure 1: C4 Level 1 — System Context for the Building Height Compliance Platform. The legend is included in the figure. Source file: `diagrams/c4-context.tex` (TikZ), version-controlled in the project repository.

External integrations

- **National Identity Provider** — used to verify that a registrant is the owner of the building being assessed. This is the control that enforces the product rule that owners may assess only their own property.
- **Google Earth Engine** — supplies the Open Buildings 2.5D Temporal rasters, which give annual building-height estimates from 2016 to 2023 [8], and building-level polygons and heights from GlobalBuildingAtlas [7]. This is the data source for the MHUC screening tier and for year-on-year growth detection.
- **MHUC GIS and Licensing Registry** — the authoritative source of zone polygons, street widths and building licence dates. Street width is the input every height rule depends on and is not available as open data.
- **Map Tile Provider** — basemap tiles for the owner and officer map views.
- **Notification Gateway** — e-mail and SMS delivery of review outcomes.

The system is split into at least two independently deployable parts: a Kotlin Multiplatform client (Android, iOS and web) and a Spring Boot backend, with the satellite screening pipeline separable as a third deployment unit because its load profile and quota constraints differ sharply from the interactive request path.

4 Functional Requirements

Fifteen user stories, prioritised with MoSCoW. The right-hand column marks what the Milestone 3 architectural prototype will implement.

Table 3: Functional requirements with MoSCoW priority and prototype scope

ID	User story	MoSCoW	In M3?
FR-01	As a *building owner*, I register and verify ownership of a specific building, so that I can assess only my own property.	Must	Yes
FR-02	As a *building owner*, I perform a guided measurement taking four elevation sightings along a walked baseline, so the system can estimate height without needing the distance to the building.	Must	Yes
FR-03	As a *building owner*, I capture a facade photograph that has faces and vehicle plates blurred and EXIF location stripped before it is stored.	Must	Yes
FR-04	As a *building owner*, I complete a measurement with no network connection and have it synchronise automatically when connectivity returns.	Should	Yes
FR-05	As a *building owner*, I receive the height as a 95 % confidence interval with the measurement tier stated, not as a single number.	Must	Yes
FR-06	As the *system*, I estimate storey count from the facade photograph using a trained vision model, as a fallback tier when sensor measurement is unavailable.	Should	Partial

ID	User story	MoSCoW	In M3?
FR-07	As the *system*, I resolve the building's zone and fronting street, and prompt the owner to confirm the frontage when GPS accuracy makes the result ambiguous.	Must	Yes
FR-08	As the *system*, I evaluate the height interval against the rule set in force on the building's licence date and return one of COMPLIANT, LIKELY_VIOLATION, INCONCLUSIVE or NEEDS_REVIEW.	Must	Yes
FR-09	As a *building owner*, I view the assessment history for my own buildings and read the decree cited by each verdict.	Must	Yes
FR-10	As an *MHUC officer*, I run a satellite screening across a district and receive a ranked list of candidate buildings to inspect.	Must	Partial
FR-11	As an *MHUC officer*, I view year-on-year satellite height changes between 2016 and 2023 to identify probable vertical extensions.	Should	No
FR-12	As an *MHUC officer*, I review a flagged assessment and record a decision with a written justification.	Must	Yes
FR-13	As an *administrator*, I author and publish a versioned, effective-dated rule set carrying a decree citation, without any code change or redeployment.	Must	Yes
FR-14	As an *administrator*, I manage user accounts and role assignments.	Should	No
FR-15	As a *building owner*, I am notified when MHUC records a decision on one of my assessments.	Could	No

Explicitly out of scope (Won't have, this release)

Legally binding compliance certificates; any enforcement or penalty workflow; payments; aerial drone imagery; and any capability for one citizen to submit an assessment of another citizen's building. The last is a deliberate product decision, not a deferral — the system is designed so that it cannot become an instrument of neighbour disputes.

5 Quality Attribute Scenarios

Six scenarios in the six-part form [1], covering six distinct quality attributes. Response measures are stated as testable thresholds.

QAS-1 *Accuracy / Functional Correctness*

Source	Building owner
Stimulus	Submits a completed four-sighting measurement with an accompanying facade photograph
Artifact	Measurement and Verdict Service (height solver, fusion, rules engine)
Environment	Normal operation; level street; device with ARCore or ARKit support; daylight
Response	The system computes a height estimate with a 95 % confidence interval, fuses the sensor and vision estimates, and issues a verdict only when the interval does not overlap the permitted height
Response measure	Height MAE $\leq$ 1.5 m against total-station ground truth over a 50-building validation set; the 95 % interval contains the true height in $\geq$ 92 % of cases; 0 verdicts of COMPLIANT or LIKELY_VIOLATION issued where the interval overlaps the limit; reduced $\chi^2 > 4$ between independent estimates routes to NEEDS_REVIEW in 100 % of cases

QAS-2 *Performance*

Source	MHUC officer
Stimulus	Requests a satellite compliance screening across an administrative district containing up to 10,000 buildings
Artifact	Satellite Screening Service and the spatial resolution component
Environment	Normal operation; satellite rasters already cached locally; 20 concurrent officer sessions
Response	The service resolves each building against its street and rule set and returns a list ranked by probability of non-compliance
Response measure	95 % of district screenings of $\leq$ 10,000 buildings complete within 90 s; interactive zone-and-street resolution for a single coordinate returns within 500 ms at the 95th percentile with 200 concurrent users; no screening exceeds the daily Earth Engine quota

QAS-3 *Availability*

Source	Building owner
Stimulus	Completes a full measurement while the backend is unreachable
Artifact	Mobile client local store, bundled rule set, and synchronisation component
Environment	Degraded mode — no network connectivity in a dense urban street
Response	The measurement is captured and persisted locally, a provisional verdict is computed from the bundled rule set and clearly marked provisional, and the record is queued for synchronisation
Response measure	100 % of offline measurements are persisted and survive a forced application close; 100 % synchronise within 60 s of connectivity returning; the capture flow completes with 0 network calls; every provisional verdict is visually marked as such and is replaced by the authoritative verdict on synchronisation

QAS-4 *Security and Privacy*

Source	An authenticated building owner, or an attacker holding a valid session token
Stimulus	Attempts to create or read an assessment for a building they have not verified ownership of
Artifact	API gateway, authorisation service, image ingestion pipeline
Environment	Normal operation
Response	The request is rejected before reaching the domain layer and the attempt is written to the audit log; separately, every ingested image is blurred and stripped of EXIF location before persistence
Response measure	100 % of cross-owner access attempts rejected with HTTP 403 across a suite of ≥ 50 automated authorisation tests; 0 unblurred images or images retaining EXIF GPS present in object storage, verified by an automated scan on 100 % of uploads; 100 % of rejected attempts appear in the audit log within 5 s

QAS-5 *Modifiability*

Source	System administrator
Stimulus	A new ministerial decree changes the permitted height bands for one zone, effective from a stated date
Artifact	Rule set store and rules engine
Environment	Production, system running and serving traffic
Response	The administrator authors a new versioned rule set with effective dates and a decree citation, publishes it, and the engine applies it to buildings licensed on or after that date while leaving earlier assessments unchanged
Response measure	Change completed in ≤ 1 person-day; 0 lines of application code modified; 0 service restarts or redeployments; 100 % of previously issued verdicts remain reproducible against the rule set version originally applied to them

QAS-6 *Usability*

Source	First-time building owner with no surveying experience
Stimulus	Attempts their first measurement of their own building
Artifact	Guided measurement flow in the mobile client
Environment	Daylight; standing on the street in front of their building; no assistance available
Response	The application computes the required walking distance in advance, guides each sighting, rejects unusable sightings with an actionable reason, and produces a valid assessment
Response measure	≥ 80 % of participants in a 10-person usability test complete a valid measurement within 4 minutes unaided; mean ≤ 2 rejected sightings per session; ≥ 90 % of participants correctly state, when asked, that the result is an estimate with a margin of error rather than an exact figure

5.1 Competing quality attributes

The brief requires at least two meaningful tensions. Four are present, and the architecture cannot satisfy both sides of any of them without an explicit decision.

Tension	Nature of the trade-off
Accuracy vs. Usability (QAS-1 vs. QAS-6)	Accuracy improves with the number of sightings and with bracing the device, but each additional sighting lengthens the flow and raises abandonment. Four sightings were chosen because they are the minimum that yields a residual usable as a fault diagnostic; two sightings have zero degrees of freedom and cannot detect their own errors.
Availability vs. Consistency (QAS-3 vs. QAS-1)	Offline capture requires a bundled rule set that may be stale, so a provisional verdict can disagree with the authoritative one. The system chooses availability for *capture* and consistency for *verdicts*: offline results are always marked provisional and are superseded on synchronisation.
Security vs. Usability (QAS-4 vs. QAS-6)	Ownership verification through an identity provider is the only robust control preventing third-party surveillance, but it adds significant registration friction before a user sees any value. Mitigation: allow an unverified trial measurement that is computed but not persisted or submitted.
Modifiability vs. Performance (QAS-5 vs. QAS-2)	Representing regulation as interpreted, effective-dated data rather than compiled code adds a lookup and evaluation step to every verdict, which is negligible for a single assessment but material across a 10,000-building screening. Mitigation: resolve and cache the applicable rule set once per zone per screening run.

6 Constraints

Table 5: Technical, business and organisational constraints

ID	Category	Constraint
CON-01	Technical	The backend must be built on Spring Boot 4.1.1 with Kotlin on JDK 21 [10], reflecting the team's existing competence and the need to share domain code with the clients.
CON-02	Technical	Android and iOS clients must be produced from a single codebase using Kotlin Multiplatform and Compose Multiplatform [11]; a five-person team cannot maintain two native codebases within the schedule.
CON-03	Technical	Spatial queries require PostgreSQL with the PostGIS extension [12]; zone containment and nearest-street resolution are not feasible without a spatial index.
CON-04	Technical	Augmented-reality ranging is platform-specific (ARCore [13] on Android, ARKit on iOS) and entirely unavailable in browsers, which expose no visual-inertial odometry. The web client therefore cannot perform sensor-based measurement.
CON-05	Technical	GNSS accuracy in dense urban streets is approximately ± 15 m, which can exceed the width of the street being classified. The architecture must treat location as uncertain rather than exact.

ID	Category	Constraint
CON-06	Technical	The system must run within a free or low-cost cloud tier for the duration of the project.
CON-07	Technical	Google Earth Engine access is quota-limited and requires a registered project; screening throughput is bounded by that quota, not by local compute.
CON-08	Business	Verdicts are advisory only. The system must not present any output as a legally binding determination of compliance.
CON-09	Business	Aerial drone imagery is excluded: UAV operation in Egypt is restricted and licensed, and surveying violations by committing one is not defensible.
CON-10	Business	Owners may assess only buildings whose ownership they have verified. Photographing and assessing third-party buildings is prohibited by product policy and enforced in the authorisation layer.
CON-11	Business	The satellite screening tier is restricted to the MHUC role and is not exposed to citizens.
CON-12	Organisational	Five-person team; academic schedule runs 19 September 2026 to 19 May 2027, with a two-week break from 16 February to 1 March 2027.
CON-13	Organisational	All deliverables must be submitted through GitHub across four milestones (M1–M4); diagram source files must be version-controlled alongside the documents.
CON-14	Organisational	Ground-truth validation depends on borrowing a total station from the Civil Engineering department; availability is outside the team's control and must be requested early.
CON-15	Data	Zone polygons and street widths are not available as open data in Egypt. Coverage is therefore limited to a single pilot district, and locations outside it must return NO_COVERAGE rather than a guessed rule.

7 Architecturally Significant Requirements

7.1 Utility tree

Each leaf is annotated **(Business Value, Architectural Impact)** on a H/M/L scale [1]. Leaves marked (H, H) are the architecture drivers.

Quality	Refinement	Scenario (abbreviated)	(BV, AI)
Accuracy	Calibrated uncertainty	Verdict withheld when the interval overlaps the limit (QAS-1)	**(H, H)**
	Cross-method agreement	Disagreeing estimates escalate to human review (QAS-1)	(H, M)
Security	Ownership scoping	Cross-owner access rejected at the gateway (QAS-4)	**(H, H)**

Quality	Refinement	Scenario (abbreviated)	(BV, AI)
	Bystander privacy	Faces and plates blurred before persistence (QAS-4)	(H, M)
Modifiability	Regulation as data	New decree deployed with no code change (QAS-5)	**(H, H)**
	Platform portability	One measurement codebase across three platforms	(M, H)
Availability	Offline capture	Measurement persists and syncs later (QAS-3)	(M, H)
	Backend degradation	Screening outage does not block owner capture	(L, M)
Performance	Screening throughput	10,000 buildings screened in 90 s (QAS-2)	(H, M)
	Interactive lookup	Zone resolution under 500 ms at p95 (QAS-2)	(M, M)
Usability	Guided capture	80 % first-time success in 4 minutes (QAS-6)	(M, L)
	Honest presentation	Users understand the result is an estimate (QAS-6)	(H, L)

7.2 The five architecturally significant requirements

ASR-1 — Calibrated uncertainty must be a first-class domain concept. (QAS-1; BV H, AI H) Every height is a distribution, not a number, and the verdict is a comparison between an interval and a threshold. This is the single most pervasive driver in the system: it dictates that the solver is a pure, independently testable component; that the database stores raw angles and baselines rather than only derived heights, so historical assessments can be recomputed when the solver improves; that the API returns intervals; and that the UI has no way to display a bare figure. Deferring this decision would require reworking every layer.

ASR-2 — Ownership scoping and bystander privacy must be structural. (QAS-4; BV H, AI H) The product is only defensible if a user cannot assess someone else's building and if third parties are never identifiable in stored imagery. Both must be enforced in the architecture rather than in the UI: authorisation sits at the gateway ahead of the domain layer, and the image pipeline blurs and strips metadata before the first write to storage, with no configuration path that disables it. Retrofitting either control would mean auditing every endpoint and every stored object.

ASR-3 — Regulation must be versioned, effective-dated data. (QAS-5; BV H, AI H) Height limits change by decree, and a building licensed in 2015 is judged by the 2015 rules. This forces a rules engine with temporal rule-set selection, immutable rule-set versions, and assessments that record which version they were evaluated against. The alternative — thresholds in code — makes the system obsolete at the next amendment and makes past verdicts irreproducible.

ASR-4 — Measurement must function fully offline. (QAS-3; BV M, AI H) Capture happens in streets with poor connectivity. The client therefore needs a local store, a bundled rule set, a synchronisation protocol, and a provisional-versus-authoritative distinction visible in the data model. Architectural impact is high even though business value is only moderate, because offline-first cannot be added later without restructuring client state management.

ASR-5 — District-scale screening must not share the interactive request path. (QAS-2; BV H, AI M) Screening ten thousand buildings against a quota-limited external API has a load profile incompatible with sub-second interactive requests. This drives separation of the screening pipeline into its own deployment unit with its own scaling and quota budget, communicating asynchronously with the core service.

Why these five

Ranking was by the product of business value and architectural impact. ASR-1 to ASR-3 score (H, H) and are non-negotiable: each one, if deferred, propagates change through every layer of the system. ASR-4 carries lower business value but equally high architectural impact, because offline-first is a structural property rather than a feature. ASR-5 is included over the remaining candidates because it is the only requirement that forces a decomposition decision — whether the system is one deployable unit or several. Usability scenarios, while valuable, are satisfied largely within a single component and therefore exert little architectural force.

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